RISC-V Instruction-Set

Erik Engheim <erik.engheim@ma.com>

Arithmetic Operation

Mnemonic	Instruction	Type	Description
ADD rd, rs1, rs2	Add	R	rd ← rs1 + rs2
SUB rd, rs1, rs2	Subtract	R	rd ← rs1 - rs2
ADDI rd, rs1, imm12	Add immediate	I	rd ← rs1 + imm12
SLT rd, rs1, rs2	Set less than	R	rd ← rs1 < rs2 ? 1 : 0
SLTI rd, rs1, imm12	Set less than immediate	I	rd ← rs1 < imm12 ? 1 : 0
SLTU rd, rs1, rs2	Set less than unsigned	R	rd ← rs1 < rs2 ? 1 : 0
SLTIU rd, rs1, imm12	Set less than immediate unsigned	I	rd ← rs1 < imm12 ? 1 : 0
LUI rd, imm20	Load upper immediate	U	rd ← imm20 << 12
AUIPC rd, imm20	Add upper immediate to PC	U	rd ← PC + imm20 << 12

Load / Store Operations

Mnemonic	Instruction	Type	Description
LD rd, imm12(rs1)	Load doubleword	I	rd ← mem[rs1 + imm12]
LW rd, imm12(rs1)	Load word	I	rd ← mem[rs1 + imm12]
LH rd, imm12(rs1)	Load halfword	I	rd ← mem[rs1 + imm12]
LB rd, imm12(rs1)	Load byte	I	rd ← mem[rs1 + imm12]
LWU rd, imm12(rs1)	Load word unsigned	I	rd ← mem[rs1 + imm12]
LHU rd, imm12(rs1)	Load halfword unsigned	I	rd ← mem[rs1 + imm12]
LBU rd, imm12(rs1)	Load byte unsigned	I	rd ← mem[rs1 + imm12]
SD rs2, imm12(rs1)	Store doubleword	S	rs2 → mem[rs1 + imm12]
SW rs2, imm12(rs1)	Store word	S	rs2(31:0) → mem[rs1 + imm12]
SH rs2, imm12(rs1)	Store halfword	S	rs2(15:0) → mem[rs1 + imm12]
SB rs2, imm12(rs1)	Store byte	S	rs2(7:0) → mem[rs1 + imm12]

Pseudo Instructions

Mnemonic	Instruction	Base instruction(s)
LI rd, imm12	Load immediate (near)	ADDI rd, zero, imm12
LI rd, imm	Load immediate (far)	LUI rd, imm[31:12] ADDI rd, rd, imm[11:0]
LA rd, sym	Load address (far)	AUIPC rd, sym[31:12] ADDI rd, rd, sym[11:0]
MV rd, rs	Copy register	ADDI rd, rs, 0
NOT rd, rs	One's complement	XORI rd, rs, -1
NEG rd, rs	Two's complement	SUB rd, zero, rs
BGT rs1, rs2, offset	Branch if rs1 > rs2	BLT rs2, rs1, offset
BLE rs1, rs2, offset	Branch if rs1 ≤ rs2	BGE rs2, rs1, offset
BGTU rs1, rs2, offset	Branch if rs1 > rs2 (unsigned)	BLTU rs2, rs1, offset
BLEU rs1, rs2, offset	Branch if rs1 ≤ rs2 (unsigned)	BGEU rs2, rs1, offset
BEQZ rs1, offset	Branch if rs1 = 0	BEQ rs1, zero, offset
BNEZ rs1, offset	Branch if rs1 ≠ 0	BNE rs1, zero, offset
BGEZ rs1, offset	Branch if rs1 ≥ 0	BGE rs1, zero, offset
BLEZ rs1, offset	Branch if rs1 ≤ 0	BGE zero, rs1, offset
BGTZ rs1, offset	Branch if rs1 > 0	BLT zero, rs1, offset
J offset	Unconditional jump	JAL zero, offset
CALL offset12	Call subroutine (near)	JALR ra, ra, offset12
CALL offset	Call subroutine (far)	AUIPC ra, offset[31:12] JALR ra, ra, offset[11:0]
RET	Return from subroutine	JALR zero, 0(ra)
NOP	No operation	ADDI zero, zero, 0

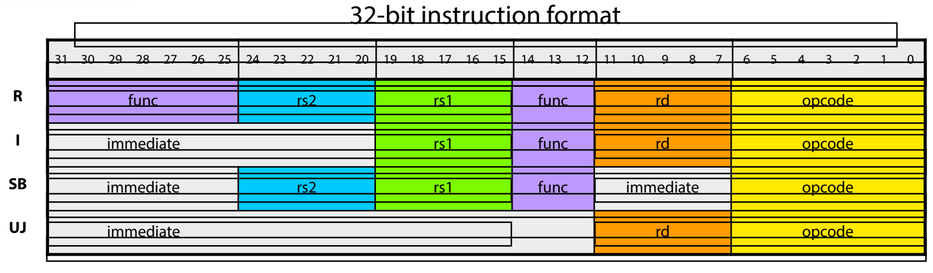
Logical Operations

Mnemonic	Instruction	Type	Description
AND rd, rs1, rs2	AND	R	rd ← rs1 & rs2
OR rd, rs1, rs2	OR	R	rd ← rs1 | rs2
XOR rd, rs1, rs2	XOR	R	rd ← rs1 ^ rs2
ANDI rd, rs1, imm12	AND immediate	I	rd ← rs1 & imm12
ORI rd, rs1, imm12	OR immediate	I	rd ← rs1 | imm12
XORI rd, rs1, imm12	XOR immediate	I	rd ← rs1 ^ imm12
SLL rd, rs1, rs2	Shift left logical	R	rd ← rs1 << rs2
SRL rd, rs1, rs2	Shift right logical	R	rd ← rs1 >> rs2
SRA rd, rs1, rs2	Shift right arithmetic	R	rd ← rs1 >> rs2
SLLI rd, rs1, shamt	Shift left logical immediate	I	rd ← rs1 << shamt
SRLI rd, rs1, shamt	Shift right logical imm.	I	rd ← rs1 >> shamt
SRAI rd, rs1, shamt	Shift right arithmetic immediate	I	rd ← rs1 >> shamt

Branching

Mnemonic	Instruction	Type	Text Description
BEQ rs1, rs2, imm12	Branch equal	SB	if rs1 = rs2 pc ← pc + imm12
BNE rs1, rs2, imm12	Branch not equal	SB	if rs1 ≠ rs2 pc ← pc + imm12
BGE rs1, rs2, imm12	Branch greater than or equal	SB	if rs1 ≥ rs2 pc ← pc + imm12
BGEU rs1, rs2, imm12	Branch greater than or equal unsigned	SB	if rs1 >= rs2 pc ← pc + imm12
BLT rs1, rs2, imm12	Branch less than	SB	if rs1 < rs2 pc ← pc + imm12
BLTU rs1, rs2, imm12	Branch less than unsigned	SB	if rs1 < rs2 pc ← pc + imm12
JAL rd, imm20	Jump and link	UJ	rd ← pc + 4 pc ← pc + imm20
JALR rd, imm12(rs1)	Jump and link register	I	rd ← pc + 4 pc ← rs1 + imm12

32-bit instruction format



Register File

x0	x1	x2	x3
x4	x5	x6	x7
x8	x9	x10	x11
x12	x13	x14	x15
x16	x17	x18	x19
x20	x21	x22	x23
x24	x25	x26	x27
x28	x29	x30	x31

Register Aliases

zero	ra	sp	gp
tp	t0	t1	t2
s0/fp	s1	a0	a1
a2	a3	a4	a5
a6	a7	s2	s3
s4	s5	s6	s7
s8	s9	s10	s11
t3	t4	t5	t6

ra - return address
sp - stack pointer
gp - global pointer
tp - thread pointer

t0 - t6 - Temporary registers
s0 - s11 - Saved by callee
a0 - a7 - Function arguments
a0 - a1 - Return value(s)